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(1) Research Program/Project/Study Title: Development of Smoked Oyster (<i>Crassostrea Iredalei</i>) in Oil	
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Program or Project has a component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation	
Municipality/City:	
Province:	
Region:	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)

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(7) Sector/Commodity	(8) Discipline/Field of Specialization
	FOOD

ABSTRACT

The composition and evaluation of a Smoke Oyster in oil are composed of the following ingredients: Oyster, the main ingredient that is locally available and rich in nutritional factors. Other ingredients includes; finger pepper, garlic, iodized salt, white pepper, bay leaves and vegetable oil. This experimental developmental research generally aims to prepare oyster smoke in oil and determine whether they are acceptable. Specifically, it sought to (1) to determine the acceptability of smoked oyster (*Crassostrea iredalei*) in oil in terms of the level of smoking time; (2) to evaluate the sensory properties of smoked oyster (*Crassostrea iredalei*) in oil in terms of taste, texture, and aroma; and (3) to assess consumer acceptability of the optimized product. Optimized product—analysis of Variance in Completely Randomized Design (CRD) to analyze data where the significant difference occurred. Using the four different time in smoking oyster, it was found out that the treatment C, which have 20 minutes of smoking time were rated " like very much smoky flavor " in terms of taste, " like very much chewy " in terms of texture, " extremely smoky aroma " in terms of aroma, " like extremely " in terms of general acceptability. The sensory quality results are subject to consumer acceptability. The employee had a rating of " like extremely." There are significant differences existing among the four treatments of smoke oyster in oil.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include the process, or methodology employed.)

Smoke seafood, smoke shells, smoke oyster in oil.

(9) Rationale:

For centuries, shellfish have been essential to culinary traditions worldwide. They can be enjoyed in numerous preparations, such as raw, steamed, grilled, or as ingredients in soups, stews, and sauces. This category encompasses two primary subgroups: crustaceans and mollusks. Molluscan bivalves such as oysters are seafoods that are widely consumed. They are

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excellent sources of protein but are highly perishable as fresh seafoods. Their short shelf life thus pose serious practical problems of their storage and distribution (*Frazier et al., 2000*).

Traditional curing methods such as sun-drying and smoking are used in preserving seafood that cannot be sold by fishermen (*Eboh et al., 2006*). Smoking was initially used for meat preservation, whereby the heat of the smoke killed surface bacteria, and regrowth was prevented by dehydration and inhibitory chemicals such as formaldehyde in wood smoke (*Potter et al., 2000*). Smoked shellfish can be savored as a delicious appetizer or snack, complemented by crackers or bread. They can also be mixed into pasta dishes, soups, and stews to provide additional depth of flavor. Furthermore, smoked shellfish make an elegant addition to seafood platters, adorned with fresh herbs and dipping sauces that elevate their rich taste.

This study intends to create a new product to increase oyster demand and acquire a distinctive and savory taste that amplifies their inherent saltiness. Overall, creating smoked oysters (*Crassostrea iredale*) gourmet can provide more customers for the farmers of oysters and innovative and appealing smoked oyster flavors for customers.

REVIEW OF LITERATURE/PRIOR ART SEARCH

CN105361017A Preparation method of canned smoked oyster

Abstract

The invention discloses a preparation method of a canned smoked oyster, and mainly solves the problems of single taste and high purine content in existing canned oyster. The method includes: selecting fresh oyster as the raw material, performing washing, then opening the shell by means of steam jet, and taking out the oyster meat; placing the oyster meat in brine to conduct ultrahigh pressure treatment, then performing washing, and dipping the oyster meat in a pre-smoking soak solution to conduct soaking so as to reduce purine; cleaning the soaked oyster meat to remove the surface salt, and performing smoking to obtain smoked oyster meat; then carrying out canning, adding soup stock, adding nisin from streptococcus lactis for bacteriostasis, carrying out exhausting and sealing, washing the canned product, performing sterilization, back pressure cooling, standing cooling to room temperature, and inspection till

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qualified, thus obtaining the canned smoked oyster finished product. The preparation method provided by the invention has the characteristics of reasonable process, strong operability and high production efficiency. The canned smoked oyster prepared according to the method has the advantages of delicious taste, unique flavor, rich nutrition, safety and health, and long shelf life.

CN105192764A Smoky instant oyster and processing method thereof

Abstract

The invention discloses a smoky flavoring processing method for instant oyster, and belongs to the technical field of seafood processing. The method comprises: taking fresh oyster as a raw material, opening the shell and fetching the meat, performing rinsing and boiling, and then performing liquid smoking, vacuum flavoring, microwave slight frying, vacuum packaging and sterilization. Because oyster possesses the characteristics of fishy smell, soft tissue, high water content and the like, tissue is easily damaged and flavoring is not easy during processing, and a conventional smoking method easily causes the problem that oyster meat shrinks because of nonuniform heat transfer and sensory quality is influenced and the like. The method employs microwave and vacuum means for promoting penetration of a smoking liquid, and solves the problems that oyster meat is difficult for liquid smoking, abdomen is easily broken during processing, water content is high, flavoring is difficult, flavoring processing time is long, processing is discontinuous and the like.

OBJECTIVES

This study aims;

1. To determine the acceptability of smoked oyster (*Crassostrea iredalei*) in oil in terms of the level of smoking time;
2. To evaluate the sensory properties of smoked oyster (*Crassostrea iredalei*) in oil in terms of taste, texture, and aroma; and
3. To assess consumer acceptability of the optimized product.

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METHODOLOGY

The study used developmental research at the Fisheries and Food Research Development Center at Capiz State University, Dayao Satellite College, Dayao, Roxas, Capiz.

The experiment was used to investigate the level of smoking time of oyster in oil. Completely Randomized Design (CRD) was utilized in the study.

Table 1.0: Proportion of ingredients

Ingredients	Quantity
Oyster (<i>Crassostrea iredalei</i>)	85 g
Finger Pepper	2g
Garlic	1g
Iodize Salt	1g
White pepper	1.50g
Bay leave	1g
Vegetable oil	50 ml

Table 1.1: Time difference during smoking of Oyster

Treatment A	Treatment B	Treatment C	Treatment D
40 mins	30 mins	20 mins	10 mins

Preparation of smoked oyster in oil

Facility

Smoke House Facility was used in this experiment.

Methods

Preparation of Oyster for Smoking

Clean the oysters and drain. Boil the the oyster for 15 minutes. Then remove. Slightly open the shells and put them into a smoke tray. Put it into a drying facility and let it dry for 1 hour. Put the dried oyster in a smokehouse facility for smoking until the taste becomes smoky.

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Preparation of Smoked Oyster in Oil

Prepare a sterilized bottle, add smoked oyster, iodize salt, garlic, white pepper, whole pepper, bay leaf, and vegetable oil. Put on a half-sealed lid. Preheat the pressure cooker and add the smoked oyster in a bottle. Pressure cook for 15 mins. in 10 pounds per square inch. Remove the bottled smoke oyster in oil from the pressure cooker and let it cool.

Scoring the sensory quality and development of smoked oyster in oil using the level of time smoking applied in oyster regarding taste, texture, aroma and general acceptability—9-point hedonic scale for the sensory characteristics and evaluated by 30 evaluators. A 9-point hedonic scale will be used for consumers' preferences, and 100 respondents will determine the products.

The data was gathered after every replication was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

RESULTS AND DISCUSSION

In terms of taste, as illustrated in Table 3.0, Treatment C (processed for 20 minutes) received the highest mean score of 7.97, corresponding to a score of 8 on the nine-point hedonic scale, which is interpreted as "Like very much, smoky flavor." In comparison, Treatment A (processed for 40 minutes) obtained the lowest mean score of 5.43, equating to a score of 5 on the same scale, which is interpreted as "Neither smoky flavor."

Table 2.0 Sensory characteristics of different treatments in terms of taste.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	5.5	5.3	5.5	16.3	5.43	Neither smoky Flavor
B	5.8	5.5	5.4	16.7	5.57	Like Slightly smoky Flavor
C	8.5	7.6	7.8	23.9	7.97	Like Very much smoky Flavor
D	7.1	6.6	7	20.7	6.90	Like Moderately smoky Flavor

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Interms of texture, as indicated in Table 2.1, Treatment C (processed for 20 minutes) achieved the highest mean score of 8.07, corresponding to a score of 8 on the nine-point hedonic scale, which translates to the verbal interpretation of "Like very much, chewy." In contrast, Treatment B (processed for 30 minutes) received the lowest mean score of 6.5, equating to a score of 6 on the same scale, which is interpreted as "Like slightly, chewy texture."

Table 2.1. Sensory characteristics of different treatments in terms of texture.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.7	7.7	7.6	22	7.33	Like Moderately chewy
B	6.5	6.3	6.7	19.5	6.5	Like Slightly chewy
C	7.7	8.3	8.2	24.2	8.07	Like Very much chewy
D	7.7	7.5	6.4	21.6	7.2	Like Moderately chewy

In terms of aroma, as presented in Table 2.2, Treatment C (processed for 20 minutes) recorded a mean score of 8.30, which equates to a score of 9 on the nine-point hedonic scale, indicating a verbal interpretation of "Extremely smoky aroma." Conversely, Treatment D (processed for 10 minutes) obtained the lowest mean score of 7.20, corresponding to a score of 7 on the same scale, which is interpreted as "Moderately smoky aroma."

Table 2.2. Sensory characteristics of different treatments in terms of Aroma.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	7.6	7.8	7.3	22.7	7.57	Very much smoky aroma
B	7.7	7.8	8.1	23.6	7.87	Very much smoky aroma
C	8.4	8.7	7.8	24.9	8.30	Extremely smoky aroma
D	7	7.4	7.2	21.6	7.20	Moderately smoky aroma

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Regarding the General Acceptability of smoked oyster in oil, as illustrated in Table 2.3, Treatment C (processed for 20 minutes) achieved the highest average score of 8.40, corresponding to a score of 9 on the nine-point hedonic scale, indicating a verbal interpretation of "like extremely." In contrast, Treatment A (processed for 40 minutes) received a mean score of 6.43, translating to a score of 7 on the same scale, which is interpreted as "like moderately."

The table displays the calculated average scores for various attributes—Taste, Texture, Aroma, Color, and Overall Acceptability—across four different processing methods, each with replication, used in the preparation of smoked oyster.

Treatment C (20 minutes) is the higher means among the four treatments regarding Taste, Texture, Aroma, and general acceptability, which implies that the formulation smoked shellfish was liked significantly by 30 panelists. The result formulation will be subjected to consumer acceptability.

Table 2.3. Sensory characteristics of different treatments in terms of General Acceptability.

Treatments	Replication			Total	Mean	Verbal Interpretation
	1	2	3			
A	6.3	6.6	6.4	19.3	6.43	Like Moderately
B	7	7.2	7.1	21.3	7.10	Like Moderately
C	8.1	8.6	8.5	25.2	8.40	Like Extremely
D	7.2	7	6.85	21.05	7.02	Like Moderately

DIFFERENCE IN THE SENSORY QUALITIES OF SMOKED OYSTER IN OIL

Descriptive statistics of the sensory characteristics in terms of taste, texture, aroma and general acceptability using 9-point hedonic scale.

In Table 3.0. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of taste.

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Table 3.0 One-way ANOVA results of the sensory characteristics in terms of taste.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	129.467	3	43.156	44.538	0.000
With groups:	112.400	116	0.969		
Total:	241.867	119			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.1. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of texture.

Table 3.1 One-way ANOVA results of the sensory characteristics in terms of texture.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	37.349	3	12.450	11.435	0.000
With groups:	125.206	115	1.089		
Total:	162.555	118			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.2. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of aroma.

Table 3.2 One-way ANOVA results of the sensory characteristics in terms of aroma.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	19.533	3	6.511	7.558	0.000
With groups:	99.933	116	0.861		
Total:	119.467	119			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

In Table 3.3. Using statistical analysis, and one-way ANOVA, results showed that there is a significant difference in the sensory characteristics in terms of General acceptability.

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Table 3.3 One-way ANOVA results of the sensory characteristics in terms of General Acceptability.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	60.733	3	20.244	32.287	0.000
With groups:	72.733	116	0.627		
Total:	133.467	119			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

CONSUMER ANALYSIS (ANOVA) RESULT

As indicated in Table 4.0. the smoked shellfish in oil received high acceptance among employee respondents, achieving a mean score of 8.44. When interpreting these scores verbally, high school students, college respondents, and employees share the same interpretation and are rated "liked extremely."

Table 4.0. The mean consumer acceptability using 9-point hedonic scale

Respondents	Total	Mean	Verbal Interpretation
Elementary Students	194	7.76	Like Very much
High School Students	204	8.16	Like Extremely
College Students	207	8.16	Like Extremely
Employee	211	8.44	Like Extremely

**Legend: 1= Dislike Extremely, 2= Dislike Very Much, 3= Dislike Moderately, 4= Dislike Slightly, 5= Neither Like nor Dislike, 6= Like Slightly, 7= Like Moderately, 8= Like Very Much, 9= Like Extremely*

In Table 4.1, using statistical analysis, and one-way ANOVA, results showed that there is no significant difference in the consumer acceptability between four group respondents.

Table 4.1 One-way ANOVA results of the sensory characteristics in terms consumer acceptability.

	Sum of Squares	df	Mean Square	F	Sig.
Between groups:	2181.618	3	727.206	0.860	0.465
With groups:	81143.382	96	845.244		
Total:	83325.000	99			

**The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.*

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CONCLUSION

- The use smoke in oysters in the preparation of smoked oysters in oil reveals that Treatment C is acceptable, earning a general rating of "liked extremely." All sensory attributes—taste, texture, and aroma—are deemed acceptable with the same rating of "liked extremely." In terms of consumer acceptability, Treatment C acceptable option among employees, boasting a high rating of 8.44. However, overall, high school students, college students, and employees have the same interpretation-rated "like extremely". This means that Treatment C is acceptable to high school students, college students, and employees and may suggests a strong potential for Treatment C in the market.
- These oil-packed smoked oyster can be enjoyed straight from the jar as a snack, incorporated into pasta dishes, or utilized as a flavorful component in various appetizers and dips.
- This product receives strong consumer result which means have potential for commercialization.

RECOMMENDATIONS

- For more interesting study results, it can be recommended that the other researchers develop a new smoked products using shellfish abundant in the Province.
- The aroma, taste, and texture of smoked oyster in oil can still be enhanced by considering other modification processes or by adding herbs and spices.
- For consumers, this study recommends consuming the smoked oyster in oil as a snack or adding into other dishies like pasta.
- Individuals who have seafood allergies should avoid this product.
- Conduct proximate analysis for this study.

(14) Output (Indicate the specific products, processes, or services that the project has produced; quantify when possible.)

Smoked Oyster in Oil

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(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

Smoked Oyster in Oil product

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI), and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.

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